STRAVYX

Australia's On-Demand Drone Services Network

Draft Data Model / ERD

First-to-accept dispatch | approvals registry | ReOC-operated docks | dagre layout (default mermaid positioning)

Stravyx Pty Ltd | Draft v0.3 (dagre) | Jul 2026 | CONFIDENTIAL

Status: Draft (v0.3) - updated per Data Model ERD Feedback v2 (Liz Ip, July 2026), checked against User Journey v5, Web App Master v6, Website Master v5, Consent & Privacy Policy v4, and Operator Database Brief Feedback v3. Supersedes v0.2.

v0.3 additions: approvals registry (operator_approvals + approval_areas), ReOC-operated dock model, assessed status + mission_feasibility, provider-agnostic payment refs (rail open - not Stripe/Xero-locked), integer cents for money. v0.2 first-to-accept dispatch is unchanged. See Section 14 changelog.

Target stack: PostgreSQL 15+ with PostGIS (geo), Redis (hot state / queues), S3 + CDN (media), AWS ap-southeast-2. Telemetry is a time-series table (TimescaleDB hypertable or native partitioning) - see Section 9.

Scope note: MVP (launch) tables are marked [MVP]; Phase 1 / Phase 2 tables are marked [P1] / [P2]. They are included now so the schema does not need destructive migrations later - build the MVP subset, leave the rest feature-flagged / unmigrated until needed.

0. How to read this document

1. Section 1 Conventions - the rules every table follows (keys, timestamps, money, soft delete, visibility).
2. Section 2 High-level map - one diagram showing the eight domains and how they connect.
3. Section 2b Dispatch mechanic - the confirmed first-to-accept flow the whole marketplace hangs off.
4. Sections 3-12 Domain sections - each has a focused ER diagram, a table-by-table breakdown with column comments, and an association rationale (why the relationship exists and its cardinality).
5. Section 13 Recommendations - things beyond the original brief worth locking (audit immutability, visibility views, race safety, retention, KYC/insurance expiry, HubSpot boundary, etc.).
6. Section 14 Changelog and Section 15 Open questions.

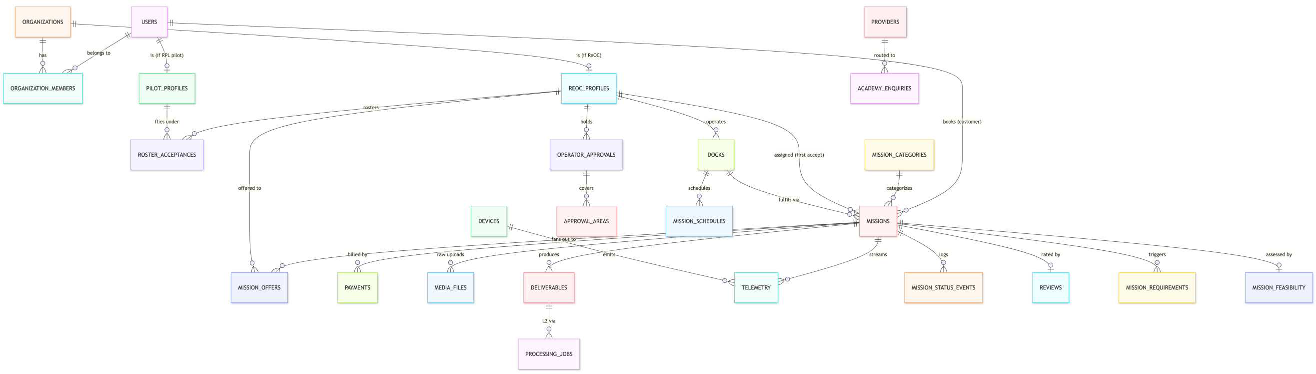
The single most important architectural constraint is the two-layer revenue model + five visibility layers (exec summary, 'Information architecture'). Money and Layer 2 data must be STRUCTURALLY unreachable by ReOCs / pilots / dock owners - not just hidden in the UI. Sensitive figures live in separate tables and I recommend DB-level projections (Section 13.2) so a leaky query cannot expose margin.

1. Conventions

Rule	Decision	Why
Primary keys	uuid (v7 preferred) id on every table	Non-guessable, shard-friendly, safe to expose in URLs; v7 is time-sortable for index locality
Timestamps	created_at, updated_at (timestamptz, UTC) on every table	Auditability; store UTC for future multi-region
Soft delete	deleted_at timestamptz NULL on user-facing entities	Marketplace disputes / regulatory audit need recoverable history; hard-delete only via retention job (13.7)
Money	bigint minor units (cents) + currency char(3) (default AUD); never floats	Decided (feedback v2) - avoids 85/15 + L1/L2 rounding drift
Payment / payout provider	Provider-agnostic provider_*_ref columns; no hard-coded rail	Capture and payout rails are open decisions - do not bake in Stripe, Xero, or any vendor until confirmed
Enums	Postgres enum for stable machine states; lookup tables for admin-editable sets	Balance integrity vs runtime flexibility (13.9)

Geo	PostGIS geography(Point,4326) / geography(Polygon,4326)	Distance for eligibility / coverage + coverage polygons
JSON	jsonb for flexible / vendor payloads (DJI raw, requirement specs)	Schema churn insulation (ADR 0001); index with GIN where queried
Idempotency	idempotency_key on externally-triggered writes (payments, uploads, HubSpot push)	At-least-once webhooks / MQTT / CRM (bridge doc 6, feedback 06)
Race safety	DB-level constraints decide contested writes, not the app	First-to-accept is a race by design - the winner is decided by a partial unique index (5, 13.14)
Audit	Append-only audit_log + status-event tables; no UPDATE/DELETE	Immutable audit log is a launch requirement
Naming	snake_case tables (plural) and columns; FK = {singular}_id	Convention consistency

2. High-level domain map



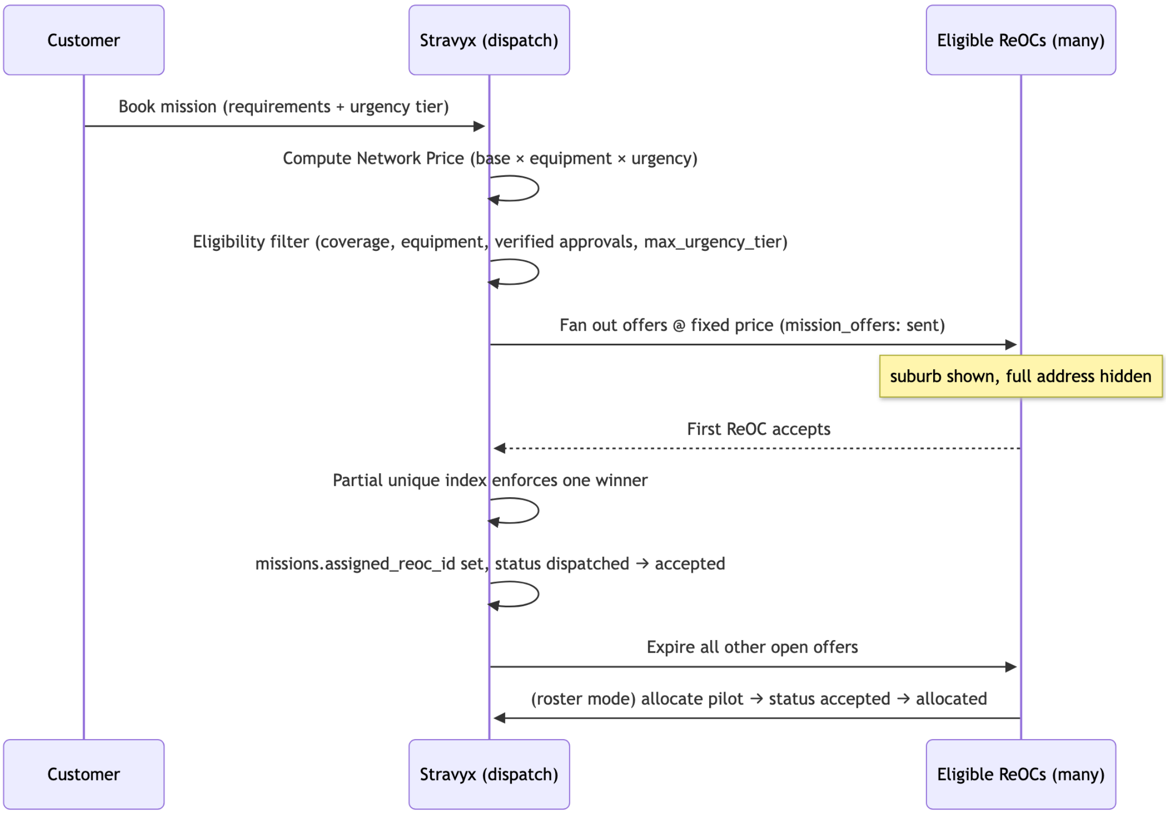
High-level domain map - the eight domains and how they connect

Eight domains (detailed sections below):

#	Domain	Sec.	Core tables	Verdict (feedback v2)
A	Identity, orgs & supply-side	3	users, reoc_profiles, pilot_profiles, roster_acceptances, operator_credentials, operator_approvals, approval_areas, service_areas, operator_equipment	Reshape - approvals registry
B	Catalogue & pricing	4	mission_categories, equipment_classes, pricing_configs, urgency_tiers	Reshape - drop price guides
C	Missions & dispatch	5	missions, mission_offers, mission_requirements, mission_feasibility, mission_schedules, ...	Reshape - dispatch + restricted-ops path
D	Payments	6	payments, payment_splits, payouts, payout_lines, refunds	Reshape - provider-agnostic; payout cardinality open
E	Media, deliverables & processing	7	media_files, deliverables, processing_jobs	Aligned
F	Devices, docks & telemetry (DJI)	8-9	devices, docks (ReOC-operated), device_bindings, telemetry, hms_alerts, livestreams	Reshape - ReOC-operated docks [MVP]; live-ops gated
G	Compliance, audit, reviews & notifications	10-11	compliance_checks, audit_log, reviews, disputes, notifications	Aligned
H	Academy & growth (CRM boundary)	12	academy_enquiries, providers (+ HubSpot mapping fields)	New - was missing

2b. The dispatch mechanic - first-to-accept (confirmed)

This is the mechanic the marketplace hangs off, confirmed by Joel (feedback 03). Read this before Domains A-C.



First-to-accept dispatch - fan-out at a fixed price, DB decides the winner

Rules baked into the schema

1. Stravyx sets the price. No bids, no floor, no ceiling, no ranking on price. The price lives on the mission; mission_offers carries NO price - there is nothing to compete on.

2. Eligibility decides who sees the offer; speed decides who wins. Fan-out targets only ReOCs that pass: service_areas (self-declared coverage), equipment match, document credentials (operator_credentials), verified CASA approval geometry covering the mission point (operator_approvals + approval_areas - NOT service_areas), and urgency-tier capability (reoc_profiles.max_urgency_tier_id).

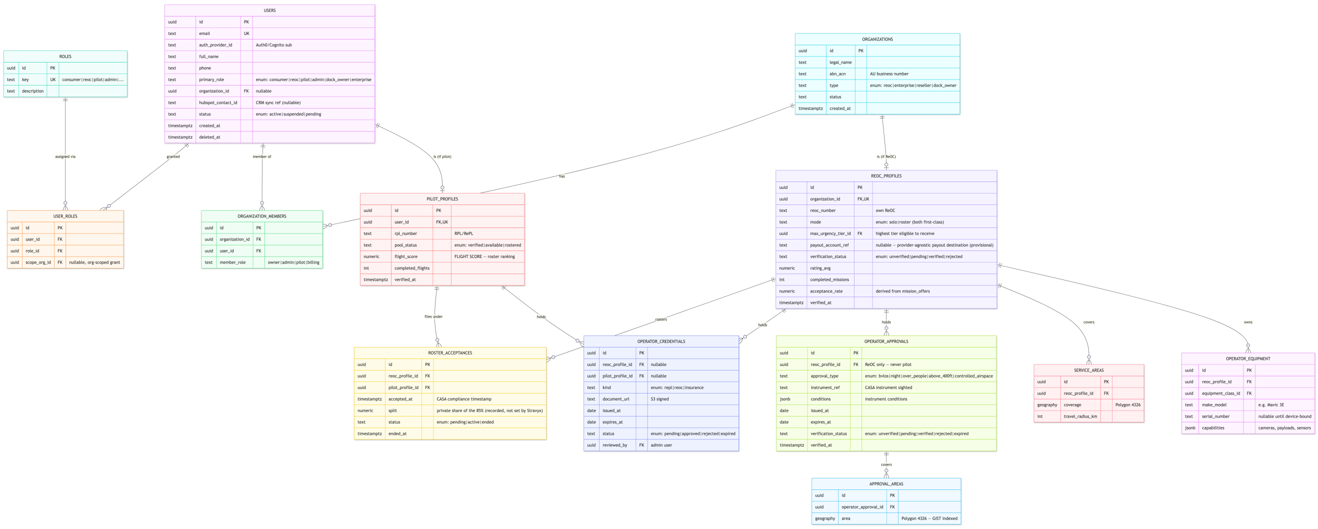
3. First to accept wins - decided in the database. A partial unique index UNIQUE (mission_id) WHERE status = 'accepted' guarantees exactly one winner under a two-tap race; the loser gets a clean 'mission already taken' (5, 13.14).

4. Offer expiry is tier-driven. expires_at = offered_at + urgency_tiers.dispatch_window. Immediate = short expiry + aggressive fan-out; Scheduled = wide window.

5. FLIGHT SCORE != dispatch ranking. FLIGHT SCORE ranks pilots in the verified pool for ReOC roster selection - it is a pilot_profiles attribute, not a bid-scoring table.

Confirmed mission state machine: booked -> dispatched -> accepted -> allocated -> assessed -> flown -> delivered (assessed = RP feasibility sign-off when required). Solo ReOC: allocated_pilot_id stays null.

3. Domain A - Identity, organizations & supply-side



Domain A - Identity, organizations & supply-side

Tables

- users - Every human / principal. primary_role is a fast-path denormalization of the RBAC in user_roles. auth_provider_id maps to Auth0/Cognito sub - no passwords stored. New: hubspot_contact_id is the one-way CRM sync ref (12 / 13.15) so the outbound push is idempotent and traceable rather than a lookup-by-email guess.
- roles / user_roles - RBAC. A user can hold multiple roles, and grants can be org-scoped (scope_org_id). This enforces the field-level visibility matrix at the API layer.
- organizations - Company entity: ReOC operators, enterprises (direct/reseller), dock owners. abn_acn supports AU compliance / invoicing.
- organization_members - Many-to-many users<->organizations with an intra-org member_role.
- reoc_profiles (was operator_profiles, split) - The DISPATCHABLE supply entity (1:1 with a reoc organization). Adds mode (solo | roster, both first-class), max_urgency_tier_id (FK to urgency_tiers; the most demanding tier this ReOC can service, applied as a hard pre-filter in fan-out), payout_account_ref (provider-agnostic settlement account for the 85% ReOC share - rail TBD), reputation, and acceptance_rate (derived from mission_offers).
- pilot_profiles (new, split out) - Verified RPL pilots in the pool, 1:1 with a user. Not directly dispatchable until rostered. pool_status (verified | available | rostered) and flight_score (FLIGHT SCORE - ranks pilots for roster selection) live here directly, not in a bid-scoring table.
- roster_acceptances (new - the missing third supply-chain entity) - The ReOC<->pilot pairing that makes a mission legally operable under CASA. Holds reoc_profile_id, pilot_profile_id, accepted_at (the compliance timestamp), split (the privately-agreed share of the 85% - Stravyx records but does not set it), and status.
- operator_credentials - RePL / ReOC / insurance docs with expires_at (critical - expiring credentials must auto-suspend eligibility, 13.5). Attaches to EITHER a reoc_profile OR a pilot_profile (exactly one FK set).
- operator_approvals (new) - Verified CASA restricted-ops approvals attached to ReOC only (never pilot). approval_type, reference_number, verified_at, expires_at.
- approval_areas (new) - PostGIS polygons for each approval. Eligibility uses ST_Covers(approval_areas.area, mission_point) - NOT service_areas.
- service_areas (was operator_service_areas) - PostGIS coverage polygons + travel radius on the ReOC. Self-declared operational coverage; separate from CASA approval geometry.
- operator_equipment - Drones the ReOC can bring (BYO). capabilities (jsonb) feeds the equipment/requirements match eligibility filter. serial_number links to a devices row once the drone streams telemetry (Section 8).

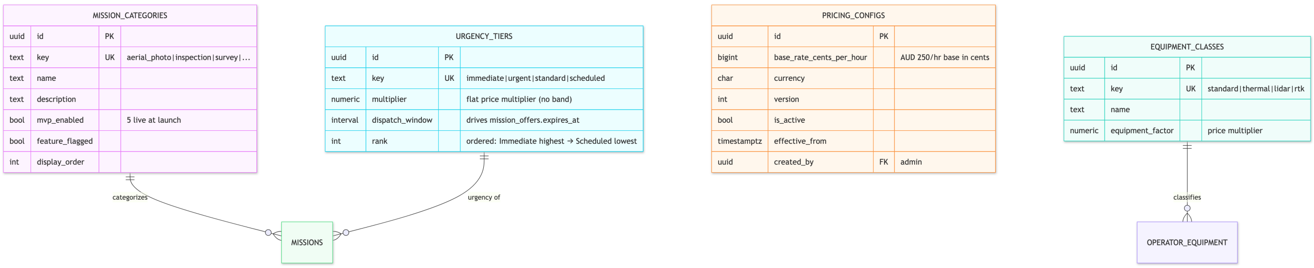
Association rationale

- organizations 1:0..1 reoc_profiles and users 1:0..1 pilot_profiles: a ReOC is an org-level dispatchable supplier; a pilot is an individual pool member. Splitting them mirrors the two distinct real-world entities and keeps dispatch logic (ReOC-only) clean from pool logic (pilot-only).

- roster_acceptances is the many-to-many-over-time bridge between ReOCs and pilots - a ReOC rosters many pilots; a pilot may fly under more than one ReOC over time. accepted_at + split make each pairing auditable for CASA.
- operator_credentials uses two nullable FKs + a check so one table serves both supply entities without a polymorphic type column.

4. Domain B - Catalogue & pricing

Reshaped (feedback B): price_guides is DROPPED - Stravyx sets the price; there is no floor, ceiling, or band. The formula tables survive and compute a flat price. urgency_tiers.dispatch_window is repurposed as the offer-expiry driver.



Domain B - Catalogue & pricing

Tables

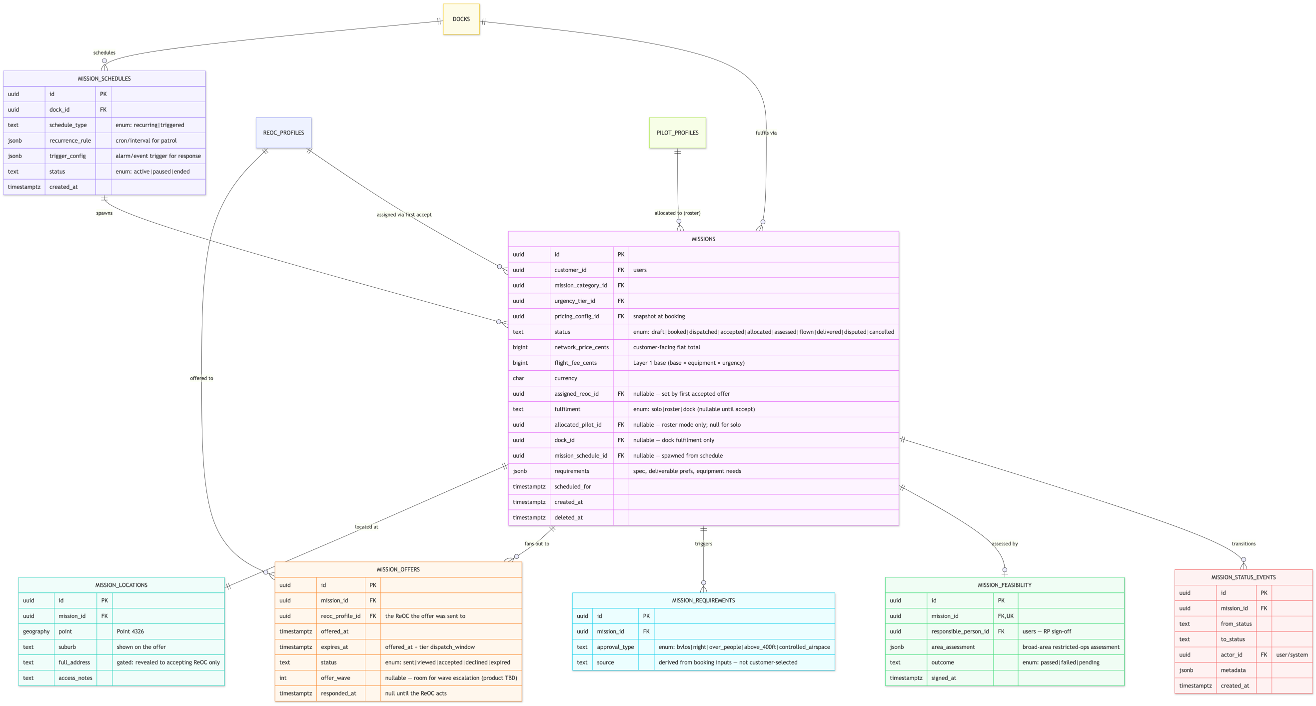
- mission_categories - The ten mission categories. mvp_enabled / feature_flagged drive '5 live at launch, remainder flagged' without code changes.
- equipment_classes - Drives the equipment factor in the price formula (base x equipment_factor x urgency). Thermal / LiDAR / RTK cost more.
- pricing_configs - Versioned, immutable base-rate config ($250/hr base). Never update a row - insert a new version and flip is_active; missions snapshot pricing_config_id at booking so historical prices are reproducible. (Feedback: 'Albert's versioned, immutable pricing pattern is exactly right - keep it.')
- urgency_tiers (reshaped) - The four tiers. multiplier is now a single flat multiplier (no min/max band, since there is no bidding). dispatch_window takes on a new job: offer expiry (mission_offers.expires_at = offered_at + dispatch_window). rank orders tiers so reoc_profiles.max_urgency_tier_id can express 'this tier and all below'.

Association rationale

- Pricing tables are referenced BY missions (not vice-versa) so historical financial records reproduce even after admins retune. price_guides is gone - there is no floor/ceiling entity because there is nothing to bid within.
- urgency_tiers now has a dual role: a pricing multiplier AND the temporal driver of offer expiry - one column (dispatch_window) cleanly produces 'Immediate = short window, Scheduled = wide window' dispatch behaviour.

5. Domain C - Missions & dispatch (the core)

Reshaped (feedback v2): adds restricted-ops path (mission_requirements, mission_feasibility, assessed status), dock fulfilment, and mission_schedules. Bids/match_scores remain dropped from v0.2.



Domain C - Missions & dispatch (the core)

Tables

- missions - State machine: draft -> booked -> dispatched -> accepted -> allocated -> assessed -> flown -> delivered (+ disputed/cancelled). assessed = RP feasibility sign-off when mission_requirements apply. Assignment on the row: assigned_reoc_id, fulfilment (solo | roster | dock), allocated_pilot_id, dock_id. Money: network_price + flight_fee (Layer 1); Layer 2 fee in payment_splits only.
- mission_requirements (new) - Restricted-ops specs (BVLOS, night, etc.) linked to mission; drives approval eligibility and assessed gate.
- mission_feasibility (new) - RP assessment record before assessed -> flown when required.
- mission_schedules (new) - Dock mission scheduling windows when fulfilment = dock.
- mission_locations - Address visibility is gated: suburb shows on the offer; full_address is revealed ONLY to the accepting ReOC. PostGIS point powers coverage eligibility and the live map.
- mission_offers (new - replaces bids) - The fan-out record: one mission, many offers, NO price (price is on the mission - nothing to compete on). expires_at is tier-driven. status (sent | viewed | accepted | declined | expired) is the offer lifecycle. Race safety (13.14): a partial unique index UNIQUE (mission_id) WHERE status = 'accepted' means the first transaction to flip an offer to accepted also sets missions.assigned_reoc_id and moves the mission dispatched -> accepted; a second concurrent tap gets a constraint failure -> clean 'mission already taken'. Also yields per-ReOC acceptance-rate and response-time.
- mission_status_events - Append-only state-transition log (event sourcing). Powers audit, the tracker UI, and dispute reconstruction. actor_id records who/what caused the transition (customer, ReOC, pilot, admin, system).

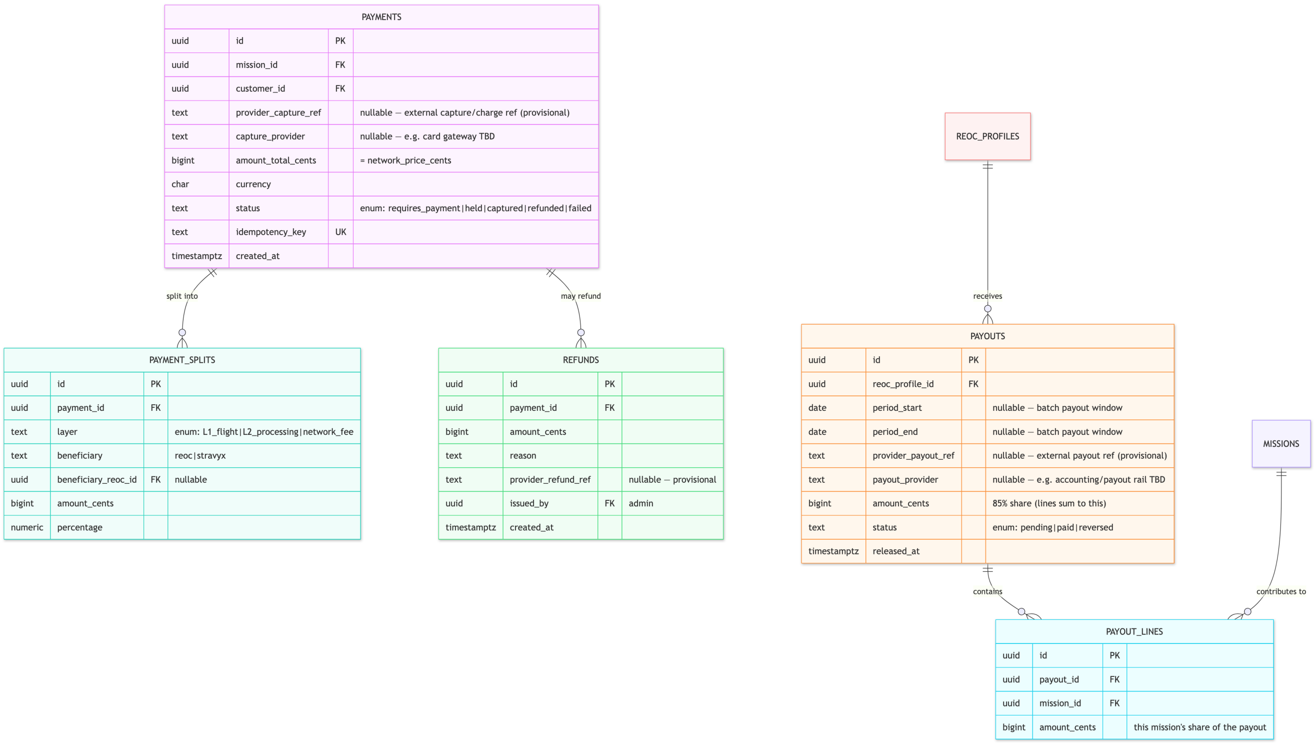
Association rationale

- missions 1:many mission_offers is the new supply-side relationship - the fan-out. There is NO bids table and NO match_scores table; nothing ranks offers because the winner is decided by speed, not score.
- reoc_profiles 1:0..1 missions via assigned_reoc_id: exactly one ReOC wins; the FK is null until first accept.
- pilot_profiles missions via allocated_pilot_id: only set in roster fulfilment, and the allocated pilot must have an active roster_acceptance with the assigned ReOC (enforce in app + FK-level check).
- missions 1:1 mission_locations, split purely for the address-visibility gate.
- mission_status_events (1:many) is immutable - append only.

FLIGHT SCORE, relocated: the retired match_scores table's purpose (ranking) does NOT apply to dispatch. FLIGHT SCORE now ranks pilots for ReOC roster selection and lives on pilot_profiles.flight_score.

6. Domain D - Payments (the margin firewall)

Reshaped (feedback v2): provider-agnostic capture/payout refs (rail open - not Stripe/Xero-locked). 85/15 split and Layer 2 firewall unchanged. payout_lines holds open payout cardinality.



Domain D - Payments (the margin firewall)

Tables

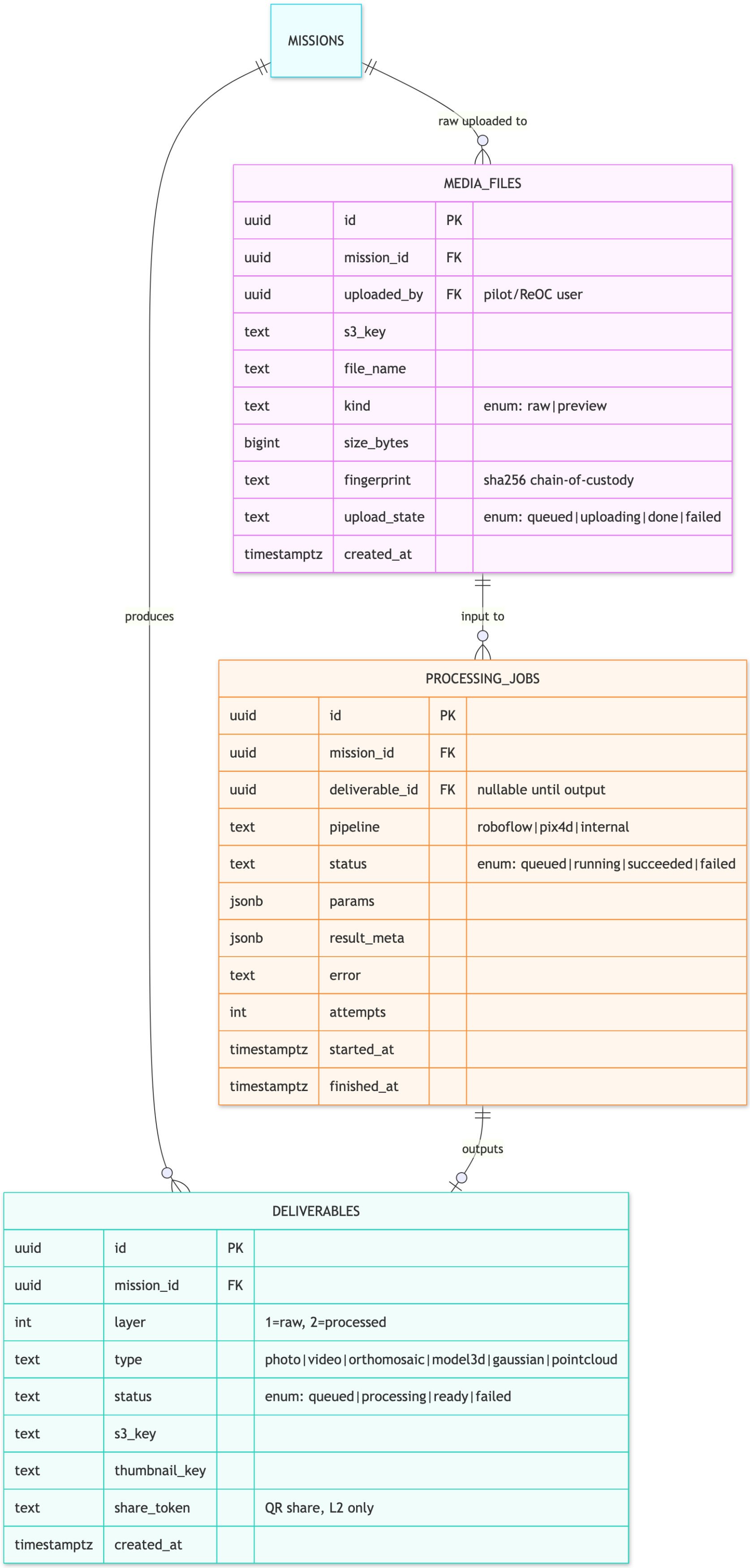
- payments - One customer charge per mission. amount_total_cents = mission network_price. provider_capture_ref + capture_provider are provisional until rail confirmed. Status: hold -> capture. idempotency_key guards duplicate webhooks.
- payment_splits - L1_flight (85% ReoC / 15% Stravyx network fee) and L2_processing (100% Stravyx). Amounts in cents (bigint).
- payouts - ReoC-facing 85% release on delivery. reoc_profile_id. provider_payout_ref + payout_provider provisional.
- payout_lines (new) - Line items linking payouts to missions; supports per-mission vs consolidated monthly payout models.
- refunds - Dispute / cancellation refunds, admin-issued, linked to the original payment.

Association rationale

- payments 1:many payment_splits: sum of splits reconciles to amount_total.
- payments 1:0..1 payouts: payout exists only after capture + release.
- Visibility enforcement: ReOCs access payouts and ONLY their own payment_splits where beneficiary='reoc' - never L2_processing / network_fee. Recommend a DB view (13.2) to make this non-bypassable.

7. Domain E - Media, deliverables & processing (Layer 2)

Aligned (feedback E): Layer 1 (raw, customer + operator visible) / Layer 2 (AI-processed, operator-invisible) split is correct. No changes.



Domain E - Media, deliverables & processing (Layer 2)

Tables

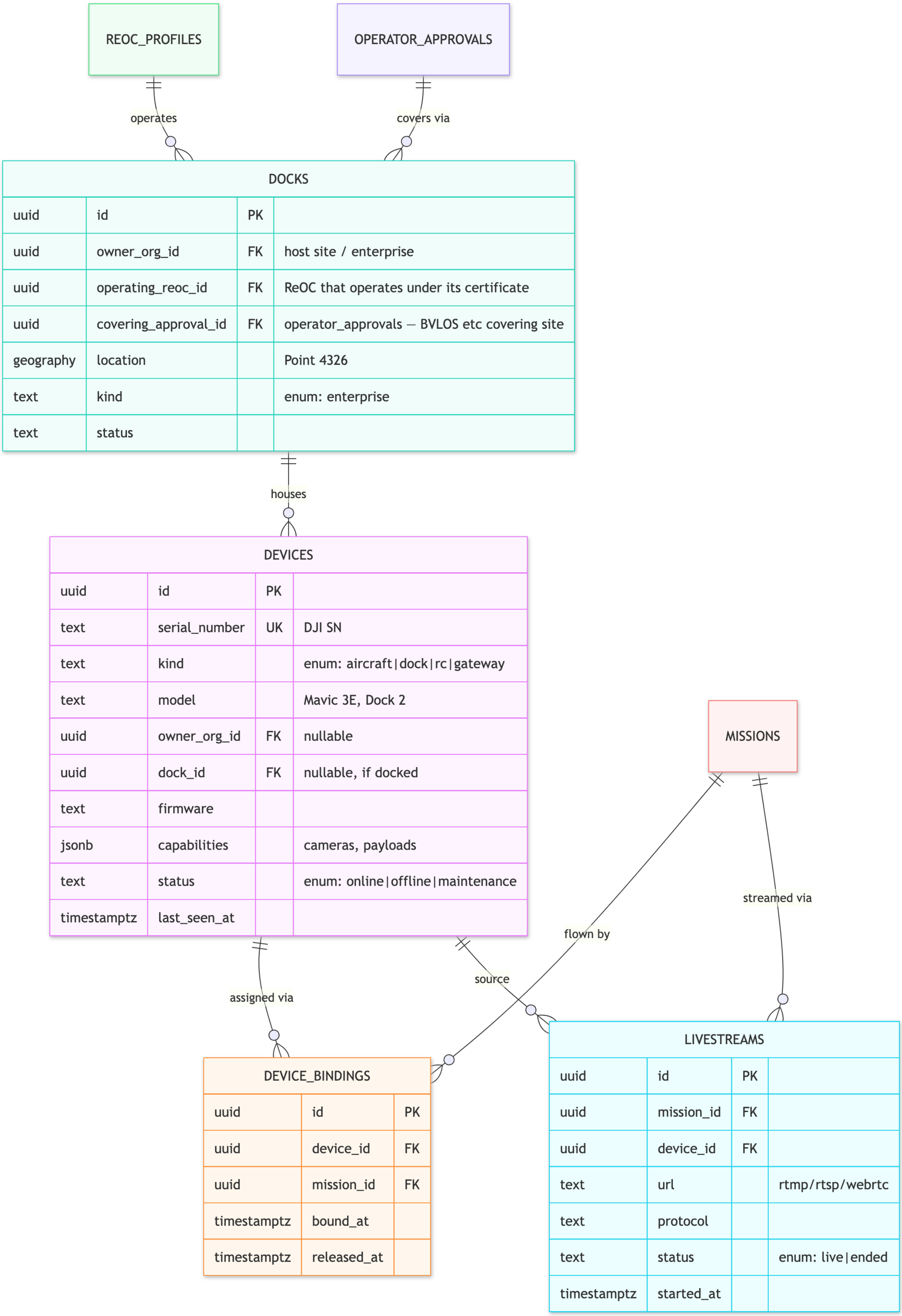
- media_files - Raw uploads to S3 (multipart via STS creds). fingerprint (sha256) = chain-of-custody for dispute audit. kind separates raw evidence from previews.
- deliverables - Layer 1 (raw, visible to customer + operator) and Layer 2 (AI-processed, operator-invisible). type matches the media viewers. share_token supports QR sharing of L2 outputs. Only s3_key stored - the API issues signed short-TTL URLs.
- processing_jobs - The Layer 2 queue. Tracks pipeline, retries, SLA timing. MVP can be a stub/manual step; ready for the AI engine (Roboflow/Pix4D) later.

Association rationale

- media_files 1:many processing_jobs 1:0..1 deliverables: processing consumes raw media and emits a processed deliverable (0..1 because a job may fail before output).
- deliverables.layer is the single switch the visibility projector uses; L2 rows are filtered out for operator roles.

8. Domain F - Devices & docks (DJI)

Reshaped (feedback v2): ReOC-operated docks [MVP]. Live-ops build gated on DJI overseas disclosure (13.16). DJI Enterprise Ecosystem application can proceed in parallel.



Domain F - Devices & docks (DJI)

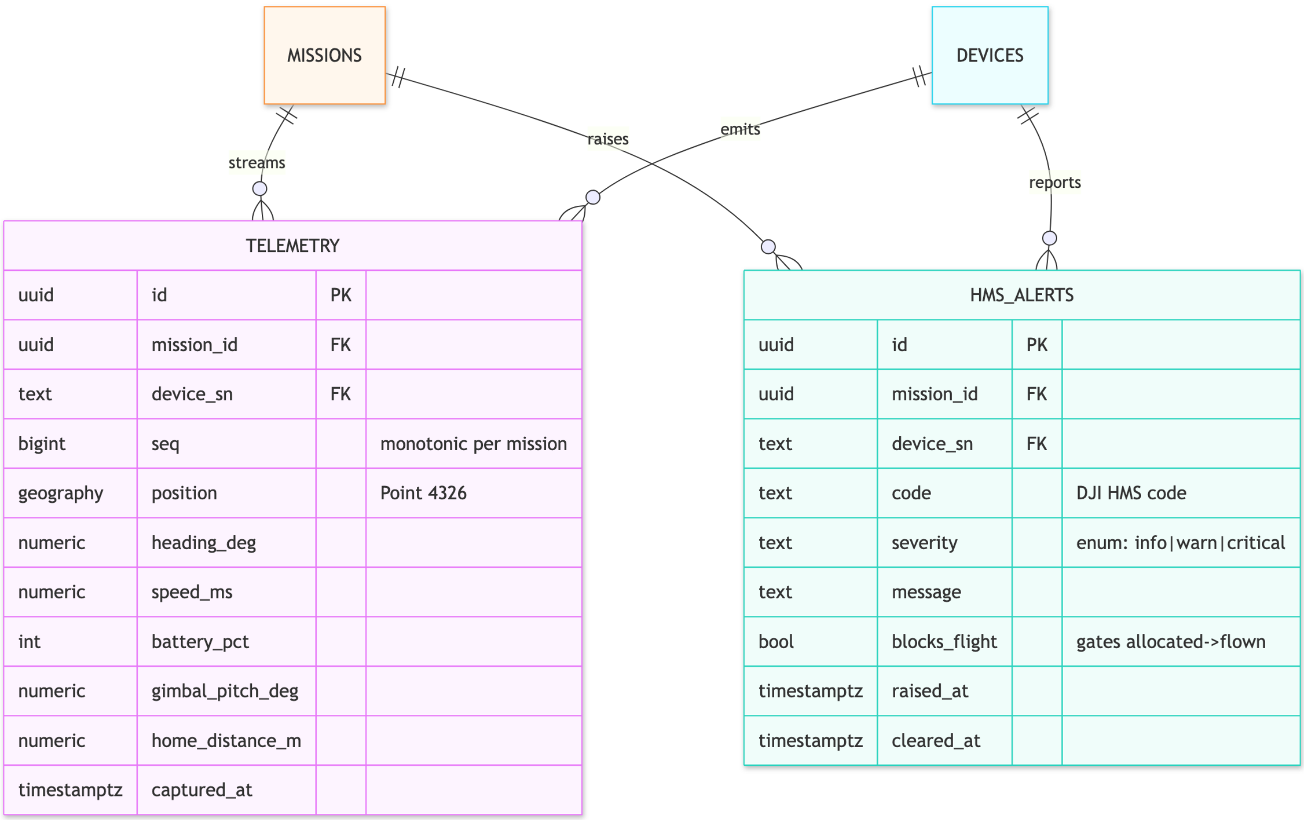
Tables

- devices - Every DJI aircraft/dock/RC/gateway keyed by serial number (the bridge's MQTT join key). capabilities verifies operator equipment claims. last_seen_at + status power the admin fleet view.
- docks [MVP] - ReOC-operated fixed DJI Dock 2/3. operating_reoc_id + covering_approval_id link dock to supply-side approvals. per_flight_fee and public_network dropped.
- device_bindings - Device<->mission mapping over time (released_at NULL = active). Backs the bridge's missionForDevice(sn) and audit ('which drone flew which mission').
- livestreams [P1] - Live video session metadata (MSDK ILiveStreamManager); media plane is RTMP/RTSP/WebRTC.

Association rationale

- docks 1:many devices; devices can also be operator-owned (nullable dock_id).
- device_bindings is many-to-many-over-time between devices and missions; released_at avoids overlapping active bindings.

9. Domain F (cont.) - Telemetry & HMS (time-series)



Domain F (cont.) - Telemetry & HMS (time-series)

Tables

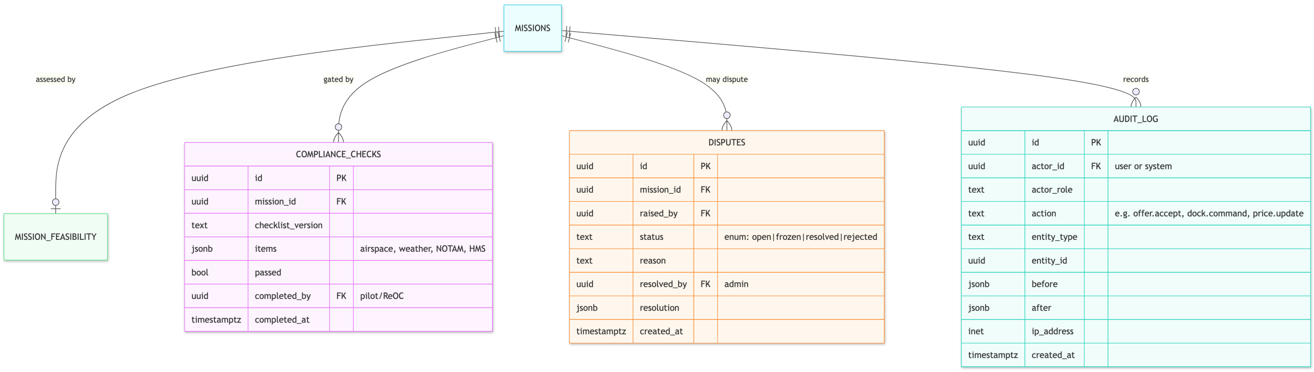
- telemetry - Normalized OSD stream (matches packages/types Telemetry, ADR 0001). High-volume, append-only, time-series. Recommendation: TimescaleDB hypertable (or native range partition on captured_at) with retention/downsampling - full-resolution for N days, then continuous aggregates for path replay. Hot last-known stays in Redis; Postgres is the durable audit/dispute path. seq gives per-mission ordering. battery_pct only - never raw cell data.
- hms_alerts - DJI Health Management System alerts. blocks_flight is the compliance gate: a critical/blocks_flight alert prevents the allocated -> flown transition (the former READY->AIRBORNE gate) and can trigger an admin freeze.

Association rationale

- Both are 1:many off missions/devices, append-only, never soft-deleted (immutable flight record). They reference device_sn (natural key) to match the SN-keyed bridge pipeline.

10. Domain G - Compliance, audit & disputes

Aligned (feedback G): immutable append-only audit log + compliance gate. No changes except mapping the gate onto the confirmed state machine.



Domain G - Compliance, audit & disputes

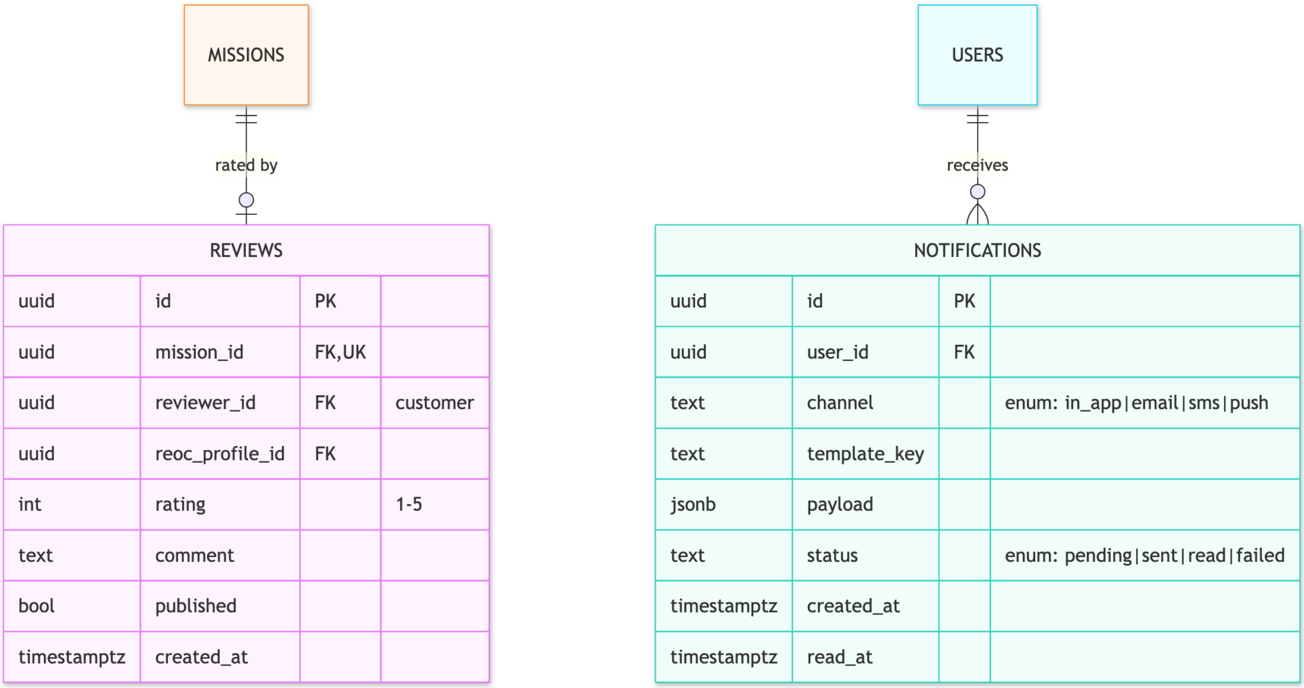
Tables

- compliance_checks - The pre-flight compliance gate, now on the allocated -> flown transition (formerly READY->AIRBORNE). items (jsonb) captures airspace/weather/NOTAM/HMS. A mission cannot go to flown without a passed check, cross-checked against hms_alerts.blocks_flight.
- audit_log - Immutable, append-only audit of every sensitive action, now including offer.accept (the race-decided assignment), dock commands, pricing changes, verification decisions, payout releases, dispute freezes. Stores before/after + ip_address. Satisfies immutable-audit + CSV-export requirements. No UPDATE/DELETE.
- disputes - Dispute lifecycle with admin freeze (halts payouts + progression). Links to refunds (Section 6).

Association rationale

- missions 1:0..1 compliance_checks (recommend 1:many to keep every attempt for audit).
- audit_log is loosely coupled (entity_type + entity_id polymorphic ref, not hard FKs) so it records actions across any table and survives archival of referenced rows.

11. Domain G (cont.) - Reviews & notifications



Domain G (cont.) - Reviews & notifications

Tables

- reviews - Customer rating (1-5) after delivered. Feeds reoc_profiles.rating_avg. 1:1 with a mission.
- notifications - Essential for a dispatch marketplace: offer-received alerts (with tier-driven urgency), offer-expiring, mission-taken, upload complete, deliverable ready, credential-expiry warnings. Multi-channel with delivery status.

12. Domain H - Academy & growth (CRM boundary)

New (feedback 05/06): the Academy domain was ABSENT from v0.1, yet POST /api/academy/enquiry is one of the three launch-hard-date endpoints (with /api/pilots/register -> pilot_profiles and /api/operators/register -> reoc_profiles). HubSpot is the engagement layer, not a second database - one-way sync only.

PROVIDERS			
uuid	id	PK	
text	name		
text	type		enum: rpa_school|training_partner|internal
text	region		
text	status		enum: active|inactive
jsonb	courses		offered programs
timestamptz	created_at		

routed to

ACADEMY_ENQUIRIES			
uuid	id	PK	
text	full_name		
text	email		
text	phone		
text	interest		course/pathway of interest
uuid	provider_id	FK	nullable — routed provider
text	source		web form / campaign
text	hubspot_contact_id		CRM sync ref (nullable)
text	idempotency_key	UK	dedupe web POST + CRM push
text	status		enum: new|contacted|converted|closed
timestamptz	created_at		

Domain H - Academy & growth (CRM boundary)

Tables

- academy_enquiries - Captures POST /api/academy/enquiry submissions (prospective pilots/students). The backend writes this record first, then pushes contact fields to HubSpot as a secondary async step. hubspot_contact_id makes the sync idempotent and traceable; idempotency_key reuses the pattern already specced for Stripe/DJI ingestion since web POST + CRM push are both at-least-once.
- providers - Training providers / RPA schools the Academy routes enquiries to. courses (jsonb) lists programs. An enquiry may be unrouted (provider_id null) until triaged.

Association rationale

- providers 1:many academy_enquiries: one provider fields many enquiries; enquiries can exist before assignment.
- CRM boundary: HubSpot is never queried for matching, dispatch, or verification, and never becomes a parallel store of licence status, mode, split, or mission state. The only touchpoints are the hubspot_contact_id mapping columns on users and academy_enquiries and the outbound idempotent push (13.15).

13. Recommendations & suggestions

13.1 Event-sourced mission state

Keep mission_status_events (and now mission_offers) as sources of truth; treat missions.status/assigned_reoc_id as projections. Free audit, time-travel for disputes, clean analytics.

13.2 Enforce visibility at the database, not just the API

The margin firewall is the business. Add Postgres views / RLS per role so a bug cannot leak margin: a ReOC view exposes flight_fee + own payment_splits(beneficiary='reoc') and omits network_price, L2_processing, network_fee; a consumer view omits operator identity pre-accept and all cost internals. Pair with contract tests.

13.3 Money as integer minor units

Consider bigint cents to avoid rounding drift on the 85/15 split and L1/L2 decomposition; reconcile sum(splits) = amount_total with a DB check.

13.4 Pricing versioning is mandatory

Never mutate pricing_configs. Missions reference the version active at booking (pricing_config_id) so historical charges are reproducible.

13.5 Credential & insurance expiry automation

operator_credentials.expires_at drives a scheduled job that auto-suspends dispatch eligibility (ReOC) or pool availability (pilot) when RePL/ReOC/insurance/restricted-ops lapse, and fires notifications ahead of expiry.

13.6 Idempotency everywhere external

Payments (Stripe), uploads (STS/multipart), MQTT ingestion, and the HubSpot push are all at-least-once. idempotency_key + dedupe on seq/event-id prevent double charges/payouts/telemetry/contacts.

13.7 Data retention & privacy (AU Privacy Act)

Retention per class: telemetry (downsample after N days), media (customer-controlled deletion, held during disputes), PII (soft-delete + scheduled hard-delete). Full address and chain-of-custody media are sensitive - document lawful basis and TTLs.

13.8 Multi-tenancy / partner readiness

Enterprise (direct/reseller) and white-label Stravyx Finance (4K Group) imply a future tenant_id/partner_id. Reserve an organization-based tenancy boundary now.

13.9 Lookup tables vs enums

Lookup tables for admin-editable sets (mission categories, urgency tiers, equipment classes); native enums for stable machine states (mission.status, mission_offers.status, payment.status).

13.10 ReOC-side reputation from offer data

mission_offers yields acceptance-rate and response-time per ReOC - a natural future input to ReOC-side reputation. reoc_profiles.acceptance_rate is a materialized rollup; keep the raw offer rows for recompute.

13.11 Geospatial from day one

PostGIS on mission_locations, service_areas, docks. Coverage eligibility in dispatch fan-out and the 'dock nodes gain structural proximity advantage' thesis depend on GIST indexing.

13.12 Feature flags in-schema

mission_categories.mvp_enabled/feature_flagged ships '5 live, 5 flagged'; a small generic feature_flags table can gate pilot/dock/enterprise portals.

13.13 Soft-delete + archival

User-facing entities soft-delete; append-only tables (audit_log, *_events, mission_offers, telemetry, hms_alerts) never delete - archive to cold storage on a retention schedule.

13.14 Race-safe first-to-accept (critical)

The winner MUST be decided by the database. Implement:

```
CREATE UNIQUE INDEX one_winner_per_mission
  ON mission_offers (mission_id)
  WHERE status = 'accepted';
```

The accepting transaction: flip the offer to accepted, set missions.assigned_reoc_id, move dispatched -> accepted, then expire remaining open offers - all atomically. The second concurrent tap hits the constraint and returns 'mission already taken.' Never resolve the winner in application code.

13.15 HubSpot - one-way sync boundary

Website forms POST to the three backend endpoints; the backend writes the record, then pushes contact fields to HubSpot as a secondary async step. hubspot_contact_id on users + academy_enquiries keeps it idempotent/traceable. HubSpot is never a source of truth for licence status, mode, split, or mission state.

13.16 DJI Cloud overseas disclosure - hard build gate

Do NOT start the live-ops build (Pilot-to-Cloud, Dock-to-Cloud, telemetry ingestion) until Consent & Privacy Policy v4 DJI overseas disclosure closes. Separate from APP 3/5 operator-database review. DJI Enterprise Ecosystem application should proceed now (approval lead time).

13.17 Restricted-ops eligibility - approval geometry (critical)

Missions with mission_requirements must match verified operator_approvals whose approval_areas cover the mission point via ST_Covers. Never substitute service_areas.

13.18 Payment / payout rail - open decision

Do not implement against a specific vendor until Liz + Joel confirm. provider_*_ref columns are provisional. payout_lines holds cardinality open for monthly consolidated ReOC payouts vs per-mission release.

14. Changelog

v0.2 -> v0.3 (Feedback v2, July 2026)

Area	v0.2	v0.3
Approvals registry	Restricted ops via operator_credentials	operator_approvals + approval_areas
Feasibility	-	mission_feasibility + assessed status

Docks	Deferred; per_flight_fee	ReOC-operated [MVP]; operating_reoc_id
Payments	Stripe-shaped fields	Provider-agnostic; payout_lines
Money	numeric(12,2)	bigint cents (decided)
DJI live-ops	Privacy-doc action	Hard build gate

v0.1 -> v0.2 (Feedback v1)

Area	v0.1 (bidding)	v0.2 (first-to-accept dispatch)
Marketplace	bids + match_scores ('BEST MATCH SCORE') + mission_assignments	Dropped. mission_offers (fan-out, no price/rank) + assignment collapsed onto missions
Winner	Highest BEST MATCH SCORE	First to accept (DB partial unique index)
Pricing	price_guides floor/ceiling; urgency = bid multiplier band	price_guides dropped; Stravyx sets flat price; urgency_tiers.dispatch_window = offer expiry
Supply profiles	Single operator_profiles	Split into reoc_profiles (mode, max_urgency_tier_id) + pilot_profiles (pool_status, flight_score)
Supply chain	-	roster_acceptances added (ReOC<->pilot, split, CASA accepted_at)
FLIGHT / BEST MATCH SCORE	Bid ranking table	FLIGHT SCORE relocated to pilot_profiles.flight_score (roster ranking)
State machine	DRAFT...QUOTING...READY->AIRBORNE... CLOSED	draft -> booked -> dispatched -> accepted -> allocated -> flown -> delivered
Compliance gate	READY -> AIRBORNE	allocated -> flown
Payments	payouts.operator_profile_id	Renamed payouts.reoc_profile_id (ReOC<->pilot split is private)
Academy	Absent	Added academy_enquiries + providers
CRM	Not modelled	hubspot_contact_id on users + academy_enquiries; one-way sync boundary
Unchanged (aligned)	-	Payments firewall, media/Layer 2, devices/telemetry (deferred), compliance/audit

15. Open questions to resolve before locking v1

Answered (feedback v2): solo ReOC leaves allocated_pilot_id null; Academy owned by this schema; integer cents; roster split standing per-pairing; enterprise billing deferred.

Still open:

1. Offer fan-out cap / wave escalation - product decision (Joel). offer_wave field reserved.
2. No-acceptance path - what happens when every offer expires? (Joel + Liz)
3. Payment / payout rail - capture provider, payout provider, fund custody, monthly vs per-mission (Liz + Joel).
4. Telemetry store - TimescaleDB vs native partitioning vs external TSDB (9). Gated behind DJI privacy gate.
5. Mission categories and urgency tier seed counts - need live source of truth (Liz + Joel).
6. max_urgency_tier_id - adopted in schema; needs Web App Master update (Liz).

Draft v0.3 - updated per Data Model ERD Feedback v2. Reconcile field names against the NestJS visibility module, live DJI payloads, and the open payment/payout rail at build time. Related: dji-frontend-technical-spec.md, dji-bridge-and-client-internals.md, executive-summary.md. Source: docs/data-model-erd.md; diagrams from docs/images/layout-dagre/rendered/.